Beyond Baseline SRL-QA: Innovations in Semantic Role-Anchored Question Answering

Algorithmic, system-level, and benchmark proposals beyond the baseline

Author: Placeholder for submission

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Abstract

This document proposes five forward-looking innovations for SRL-based question answering. The proposals are motivated by the empirical limitations of the baseline PropQA-Net system and by recent literature on QA-based semantics, PropBank adjudication, and frame-aware LLM adaptation. The central thesis is that classical SRL-QA becomes substantially more useful when it is extended with frame memory, cross-sentence reasoning, calibrated answer selection, and workflow automation rather than treated as a static span-labeling problem.

Introduction - Why Innovate?

The empirical analysis shows that a classical BiLSTM-based SRL-QA system is workable yet structurally limited. It recovers many local arguments but struggles with rare roles, discontinuous spans, and questions whose answer depends on more than one predicate frame. Recent 2022-2025 literature points in the same direction: question-based semantics is expanding into richer representation spaces, and LLM-era work is rediscovering the value of explicit semantic structure for reliable reasoning.

The innovation agenda pursued here therefore targets five fronts: better predicate disambiguation, multi-sentence reasoning, multi-predicate answer composition, calibrated abstention, system-level orchestration, and a benchmark culture that can evaluate these capabilities directly. The goal is not to abandon classical SRL-QA, but to make it a stronger substrate for research and deployment.

Innovation 1 - Algorithmic: PropQA-Net Architecture Enhancements

Frame-aware predicate disambiguation extends PropQA-Net by replacing isolated predicate embeddings with a roleset memory. Each candidate roleset contributes key-value pairs encoding role descriptions, example glosses, and optional lexical paraphrases. During decoding, the model attends over this memory so that ambiguous predicates such as 'charge', 'leave', or 'hold' can be resolved under soft semantic constraints rather than surface context alone.

Semantic frame chaining addresses cross-sentence QA by representing each predicate instance as a node in a document-level graph and linking nodes through shared arguments, coreference, temporal adjacency, or discourse cues. The answer mechanism then traverses the graph from the predicate most directly evoked by the question toward supporting frames, allowing a question to combine evidence from event causes, consequences, or follow-up actions.

Multi-predicate answer composition adds a second-stage span-composition layer that aggregates compatible spans from multiple predicates with attention weights normalized over question relevance, frame confidence, and entity consistency. Confidence-calibrated answer selection replaces hard argmax decoding with a distribution over candidate spans and introduces an explicit abstain option when no span exceeds a calibrated confidence threshold.

Taken together, these enhancements redefine PropQA-Net from a sentence-local tagger into a semantic retrieval-and-reasoning system. The crucial shift is that rolesets, spans, and frame links become memory

structures rather than passive annotations. That creates a pathway from classical SRL toward richer semantic reasoning without discarding explicit role structure.

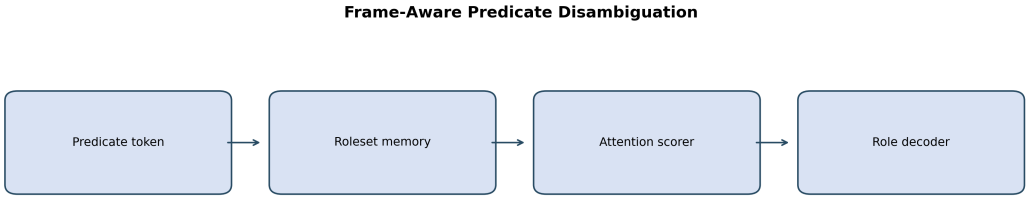

Figure 1. Frame-aware predicate disambiguation with roleset memory.

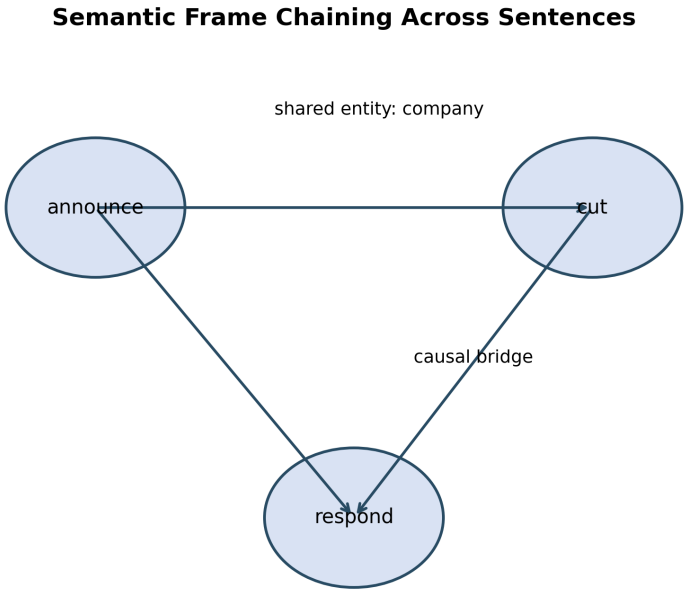

Figure 2. Semantic frame chaining for multi-sentence reasoning.

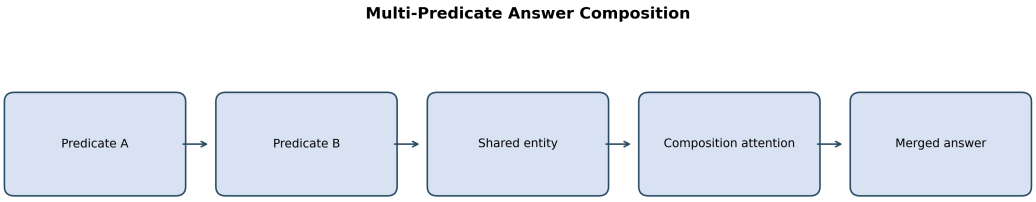

Figure 3. Multi-predicate answer composition.

Innovation 2 - System-Level: Automated SRL-QA Workflow

The proposed PropAgent pipeline decomposes SRL-QA into cooperating agents: IngestAgent, ParseAgent, SRLAgent, IndexAgent, MapperAgent, ReasonerAgent, ValidatorAgent, and ReportAgent. This architecture makes the workflow inspectable and fault-tolerant. Parsing or SRL failures do not immediately collapse the entire system; instead, downstream agents can trigger repair logic, fallback heuristics, or selective abstention.

Inter-agent communication should be structured around a shared proposition store indexed by document span, predicate identifier, roleset, argument role, entity links, and confidence. MapperAgent interprets the question against this store, ReasonerAgent performs frame traversal or composition, ValidatorAgent checks negation, modality, and scope, and ReportAgent verbalizes the final answer with an explanation trail. Such a pipeline scales more naturally to enterprise document collections than a monolithic end-to-end reader.

Failure recovery is a first-class requirement in this design. If ParseAgent produces weak syntax, SRLAgent can still attempt predicate-local extraction with lower confidence; if SRLAgent fails, MapperAgent can request a shallow fallback using lexical cues; if ReasonerAgent detects incompatible evidence, ValidatorAgent can trigger abstention rather than forced answering. This explicit recovery policy is essential for trustworthy document analytics.

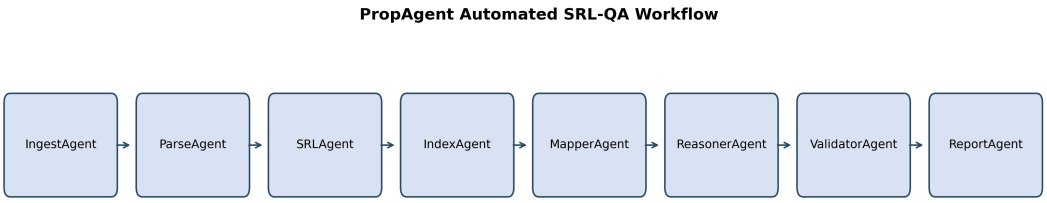

Figure 4. PropAgent system pipeline.

Innovation 3 - Application-Level: Novel Use Case Designs

Legal contract QA benefits from role-grounded obligations because obligations, beneficiaries, deliverables, deadlines, and conditions map naturally onto PropBank-style slots. Medical note QA can use the same abstraction to tie medication administration, dosage, recipient, and rationale into a single event schema. Safety incident reports, financial disclosures, and procedural manuals each share the same advantage: the critical question is often not what text overlaps with the query, but which participant filled which role in which event.

Across these domains, SRL can outperform generic QA as an organizing layer because it exposes event structure explicitly. A contract reviewer asking who must deliver what by when, or a clinician asking what medication was administered to whom and for what reason, is effectively posing a role-targeted question. This makes SRL-QA especially appealing for applications requiring explanation, regulation, or auditability.

A concrete legal example is a clause such as 'The supplier shall deliver the replacement units to the buyer within ten business days after notice.' An SRL-QA system can ground supplier as ARG0, replacement units as ARG1, buyer as ARG2 or destination-like role, and the deadline as ARGM-TMP. A generic QA system may retrieve the same text span, but it will not necessarily expose the event schema supporting the answer.

A clinical example is 'The nurse administered 5 mg of morphine to the patient for severe pain.' Here medication, dosage, recipient, and rationale are all role-bearing spans that matter operationally. The same structured pattern recurs in incident analysis, securities filings, and procedural manuals, which suggests

that SRL-QA has unusually broad application leverage relative to its modest modeling footprint.

Innovation 4 - Research Direction: SRL-QA + LLM Hybrid Architecture

HybridPropQA uses a classical SRL parser as the structural front end and an LLM as a soft reasoner and verbalizer. The parser extracts predicate-argument candidates and roleset constraints; the LLM then reasons over those structured candidates under prompt constraints that mention allowable roles, confidence scores, and abstention policy. In this design, the LLM does not replace symbolic structure but consumes it.

The expected benefit is asymmetry: classical SRL contributes consistency and span faithfulness, while the LLM contributes paraphrase handling, broader world knowledge, and natural-language explanation. Roleset grounding is the critical safeguard because it limits the degrees of freedom available to the generator and keeps answers tied to an explicit semantic inventory.

This hybrid direction also offers a practical research compromise. It avoids the unrealistic expectation that small academic projects must retrain giant models end to end, while still enabling the use of advanced language models where they are strongest: explanation, paraphrase tolerance, and multi-step reasoning over already structured evidence.

In the upgraded local implementation, this design is realized as a deterministic hybrid inference layer rather than as a generative replacement model. PropQA-Net remains the baseline SRL-QA backbone, while role-aware heuristics, transformer QA spans, semantic reranking, and concise reasoning traces are combined into a research-facing system that stays reproducible on commodity hardware.

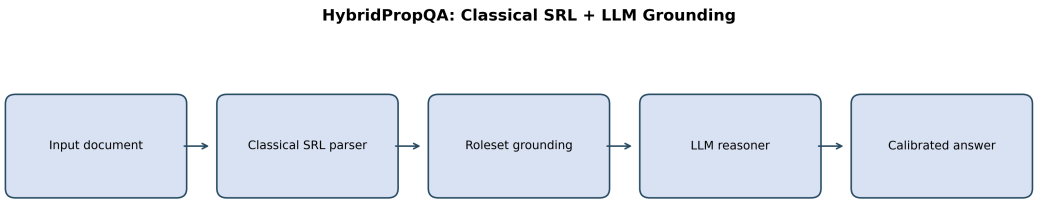

Figure 5. HybridPropQA architecture.

Innovation 5 - Benchmark Proposal

PropQA-Bench is proposed as a benchmark for predicate-local, multi-predicate, and cross-sentence SRL-QA. Annotation would begin with document sampling and predicate selection, followed by roleset-aware question writing, expert answer adjudication, and release of both extractive gold spans and supporting semantic graphs. Evaluation would report exact match, token F1, calibrated abstention, role accuracy, graph-traversal success, and explanation faithfulness.

The benchmark should include classical baselines, neural SRL baselines, retriever-reader baselines, and hybrid classical-LLM systems. A public leaderboard would separate local argument questions from compositional multi-hop questions so that systems are rewarded for transparent semantic reasoning rather

than only lexical overlap.

The broader value of such a benchmark is cultural as well as technical. It would give the community a shared target for questions that are too structured for open-domain QA benchmarks and too compositional for sentence-local SRL corpora. In doing so, it could align future work on dataset design, calibration, and explanation around a common evaluation language.

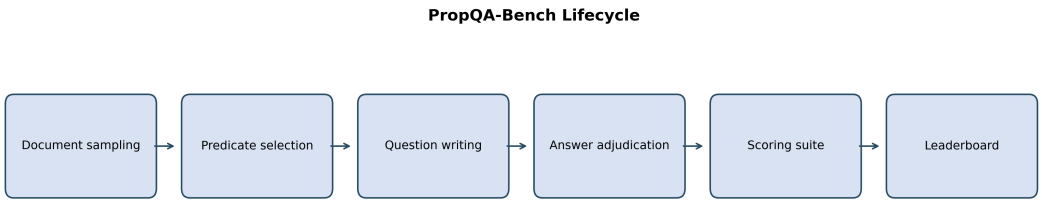

Figure 6. PropQA-Bench lifecycle.

Conclusion and Research Roadmap

A practical roadmap would prioritize calibrated answer selection first, frame-aware disambiguation second, and document-level frame chaining third. Those three steps directly attack the largest empirical weaknesses exposed by the baseline. Multi-predicate composition and full PropAgent orchestration can then be layered on top as scaling mechanisms.

Over a three-year horizon, SRL-QA research should aim to produce benchmarks that reward explanation, multilingual resources that extend QA-based semantics beyond English, and hybrid architectures that preserve the interpretability of classical semantics while leveraging LLMs for broader reasoning and interaction.

The overarching message is that innovation in SRL-QA should not chase raw model size alone. The most promising path is structured compositionality: better use of rolesets, better links between events, better calibration, and better workflows for turning semantic structure into dependable answers.

Implementation Sketches

A first implementation of frame-aware disambiguation would precompute candidate rolesets per predicate, encode their textual role descriptions, and score them jointly with the contextual predicate representation before BIO decoding. This is a lightweight upgrade path because the required frame files already exist in PropBank.

A first implementation of PropAgent would serialize every detected predicate into a proposition index with document position, roleset, arguments, and confidence. Downstream agents could then operate over that index without rerunning the whole parser, which is essential for scalable document workflows.

A first implementation of PropQA-Bench would focus on specification: task templates, adjudication policy, and metric definitions. Establishing those conventions is the prerequisite for any meaningful community benchmark.

Use-Case Walkthroughs

In a legal-contract workflow, PropAgent would first index obligations and deliverables, then answer questions such as who is obligated to deliver what by when under which conditions. The value of SRL here is that the answer must remain tied to a clause and role structure that auditors can inspect.

In a clinical-note workflow, the same pipeline could link medication events across progress notes, preserving the mapping between medication, recipient, dosage, timing, and reason. This is a natural fit for calibrated abstention because uncertainty is itself clinically meaningful.

In safety-incident analysis, frame chaining becomes particularly valuable because cause, response, and outcome often appear in separate sentences. A benchmark that only rewards local argument spans would miss the most decision-relevant questions in this domain.

Financial-disclosure QA introduces another twist: conditions and counterfactual clauses often matter as much as the core acquisition event. Role-aware reasoning can separate acquisition price, acquirer, acquired entity, and contingent conditions more reliably than overlap-driven QA alone.

Instructional-procedure QA highlights temporal and prerequisite reasoning. Here SRL-QA can expose not only what must be done, but before which step, by whom, and for what purpose. That combination makes the domain a strong candidate for early PropQA-Bench pilots.

Risks and Mitigations

The main risk in these innovation proposals is overcomplicating a baseline that is valuable precisely because it is transparent. The mitigation is to stage improvements so that every new component exposes an interpretable intermediate artifact: roleset memory scores, graph traversals, calibration thresholds, or validator decisions.

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